

Supplementary Information for

Chiral Europium Halides with High-Performance Magnetic Field Tunable Red Circularly Polarized Luminescence at Room Temperature

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The PDF file includes:

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Supplementary references

Supplementary Figures

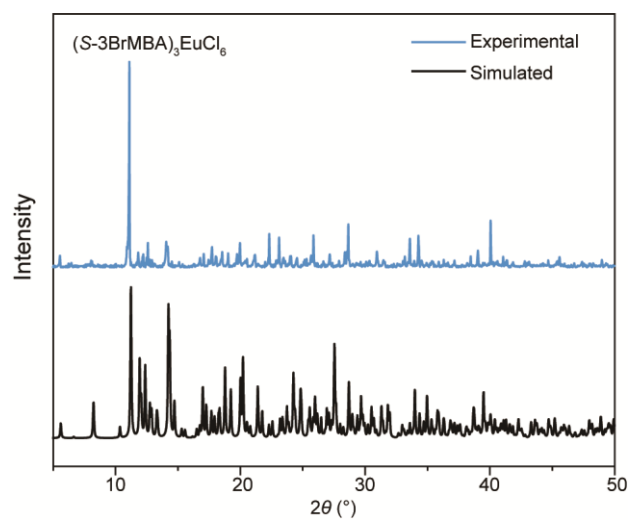


PLATON/ADDSYM for r-eu P 21 21 21													
ADDSYM Search on ALL NON-H Chemical Types [Max NonFlt 0 Perc]													
Density based on Input Atom Set = 0.685 g.cm-3 - Vol / Non-H atom =126.2 Ang+3													
Criteria 0.30 Deg (Metric), 0.25 Ang (Rot), 0.25 Ang (Inv), 0.25 Ang (Transl)													
Symm.	Input	Reduced	(Ang)	(Deg)	Perc	AvrDev.	(Ang)	Input Cell					
Elem	Cell_Row	Cell_Row	d	Typ	Dot	Angle	Flt	MaxDev.	x	y	z		
2	[1 0 0]	[1 0 0]	7.65	2	1	0	100	0 Through	0	1/4	1/2		
1								0 Screw	1/2	0	0		
2	[0 1 0]	[0-1 0]	14.62	2	1	0	100	0 Through	1/2	0	1/4		
1								0 Screw	0	1/2	0		
2	[0 0 1]	[0 0 1]	31.58	2	1	0	100	0 Through	1/4	1/2	0		
1								0 Screw	0	0	1/2		
Reduced-to-Convent			Input-to-Reduced			T = Input-to-Convent:			a' = T a				
(	1	0	0)	(	1	0	0)	(	1	0	0)	Det(T)	
(	0	-1	0)	X	(	0	-1	0)	(	0	1	0)	=
(	0	0	-1)	(	0	0	-1)	(	0	0	1)	1.000	
Cell Lattice	a	b	c	Alpha	Beta	Gamma	Volume	CrystalSystem	Laue				
Input aP	7.655	14.621	31.584	90.00	90.00	90.00	3535	orthorhombic	mmm				
Reduced P	7.655	14.621	31.584	90.00	90.00	90.00	3535						
Convent aP	7.655	14.621	31.584	90.00	90.00	90.00	3535	orthorhombic	mmm				
:: Origin Shifted to: 0.500, 0.500, 0.500 after Cell Transformation													
:: SpaceGroup = P212121 - No Obvious Spacegroup Change Needed/Suggested													

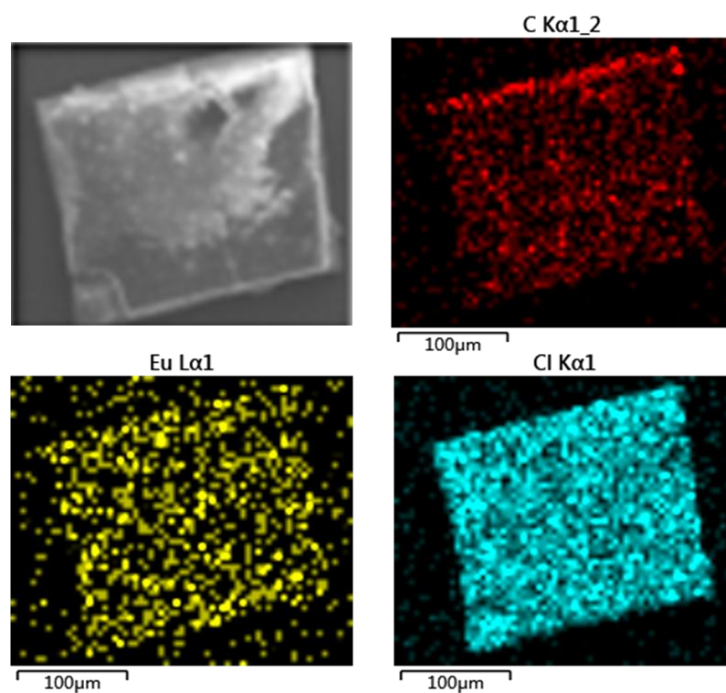


PLATON/ADDSYM for s-eu													P 21 21 21				
ADDSYM Search on ALL NON-H Chemical Types [Max NonFlt 0 Perc]																	
Density based on Input Atom Set = 0.684 g.cm ⁻³ - Vol / Non-H atom =126.4 Ang+3																	
Criteria 0.30 Deg (Metric), 0.25 Ang (Rot), 0.25 Ang (Inv), 0.25 Ang (Transl)																	
Symm.		Input		Reduced		(Ang)		(Deg)		Perc		AvrDev.		(Ang)		Input Cell	
Elem	Cell_Row	Cell_Row	d	Typ	Dot	Angle	Flt	MaxDev.				x	y	z			
2	[1 0 0]	[1 0 0]	7.65	2	1	0	100	0	Through	0	3/4	1/2					
1								0	Screw	1/2	0	0					
2	[0 1 0]	[0-1 0]	14.65	2	1	0	100	0	Through	1/2	0	3/4					
1								0	Screw	0	1/2	0					
2	[0 0 1]	[0 0 1]	31.58	2	1	0	100	0	Through	3/4	1/2	0					
1								0	Screw	0	0	1/2					
Reduced-to-Convent			Input-to-Reduced			T = Input-to-Convent:			a' = T a								
(	1	0	0)	(	1	0	0)	(	1	0	0)	Det(T)					
(	0	-1	0)	X	(	0	-1	0)	(	0	1	0)	=				
(	0	0	-1)	(	0	0	-1)	(	0	0	1)	1.000					
Cell Lattice	a	b	c	Alpha	Beta	Gamma	Volume	CrystalSystem	Laue								
Input aP	7.651	14.653	31.580	90.00	90.00	90.00	3540	orthorhombic	mmm								
Reduced P	7.651	14.653	31.580	90.00	90.00	90.00	3540										
Convent aP	7.651	14.653	31.580	90.00	90.00	90.00	3540	orthorhombic	mmm								
:: Origin Shifted to: 0.500, 0.000, 0.500 after Cell Transformation																	
:: SpaceGroup = P212121 - No Obvious Spacegroup Change Needed/Suggested																	

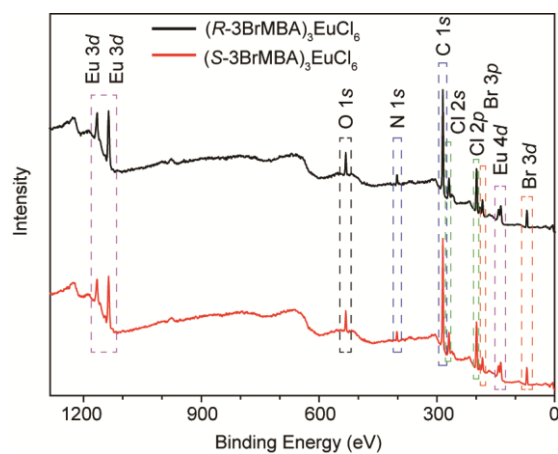
Supplementary Fig. 1 | Post-refinement analysis for missing symmetry using PLATON's ADDSYM tool on isolated inorganic frameworks in (*R/S*-3BrMBA)₃EuCl₆. All the C, H, and N atoms were manually deleted from the fully refined structure using the CrystalMaker software. The remaining europium halide framework is exported into a CIF file for PLATON symmetry analysis. Default values of angle criterium (i.e., 0.3° for lattice metrical symmetry) and distance criterium (i.e., 0.25 Å for coinciding atoms for rotational, inversion, and translational symmetry elements) were used in the symmetry search. There is no change from the original *P*2₁2₁2₁ space group.



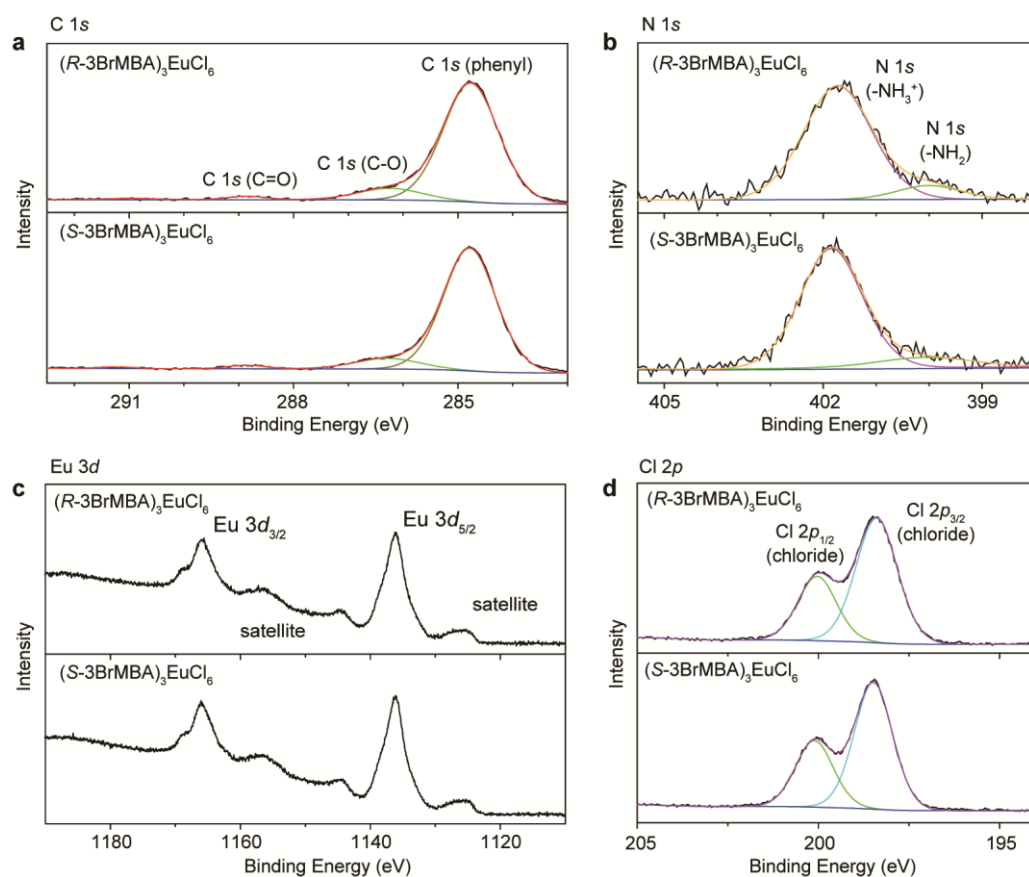
Supplementary Fig. 2 | The powder X-ray diffraction (PXRD) patterns of $(S-3BrMBA)_3EuCl_6$. The blue line is the experimental results, and the black line is the corresponding simulated results from the crystal structure.



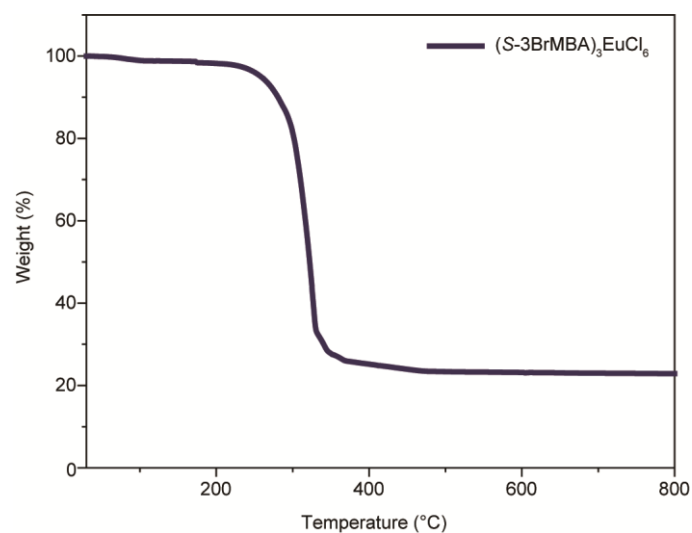
Supplementary Fig. 3 | Scanning electron microscopy (SEM) and energy dispersive spectrometer (EDS) mapping images for $(S\text{-}3\text{BrMBA})_3\text{EuCl}_6$. SEM image of $(S\text{-}3\text{BrMBA})_3\text{EuCl}_6$ microcrystal and EDS mapping images of C (red), Eu (yellow), and Cl (blue) distribution.



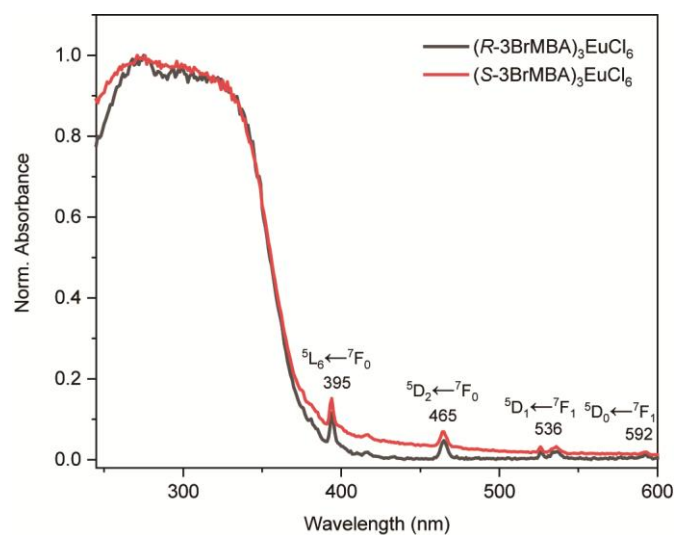
Supplementary Fig. 4 | X-ray photoelectron spectroscopy (XPS) survey spectra for chiral europium halides. The XPS survey spectra of $(R-3BrMBA)_3EuCl_6$ (black) and $(S-3BrMBA)_3EuCl_6$ (red).



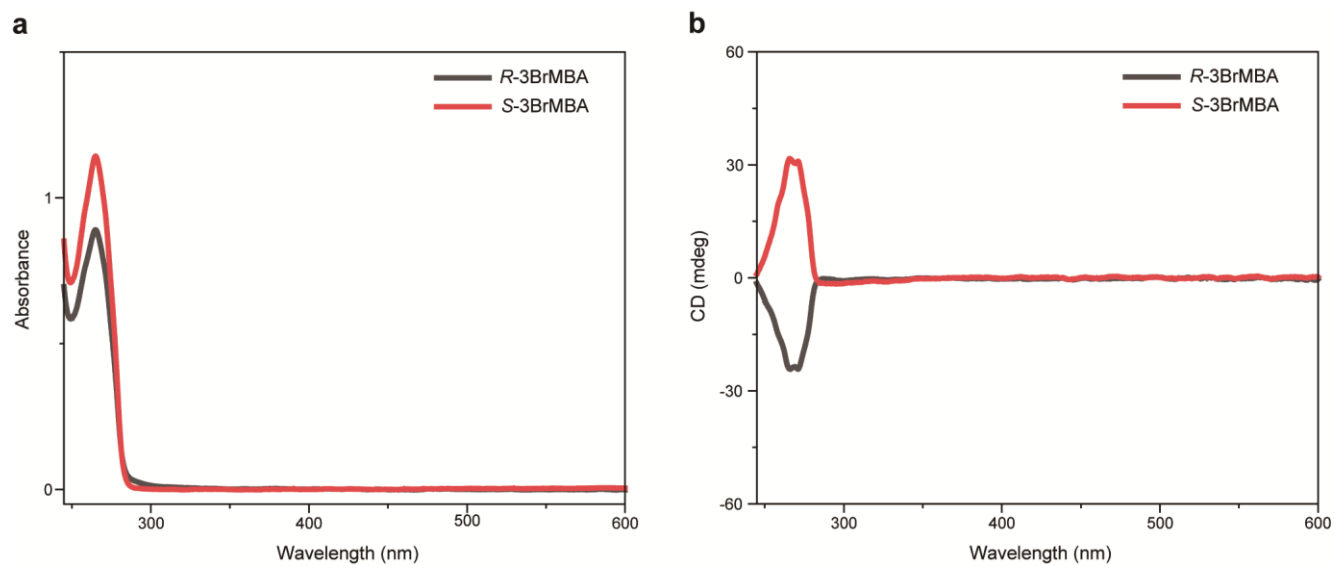
Supplementary Fig. 5 | XPS spectra of elements in chiral europium halides. The XPS spectra of C 1s (a), N 1s (b), Eu 3d (c), and Cl 2p (d) in $(R-3BrMBA)_3EuCl_6$ and $(S-3BrMBA)_3EuCl_6$.



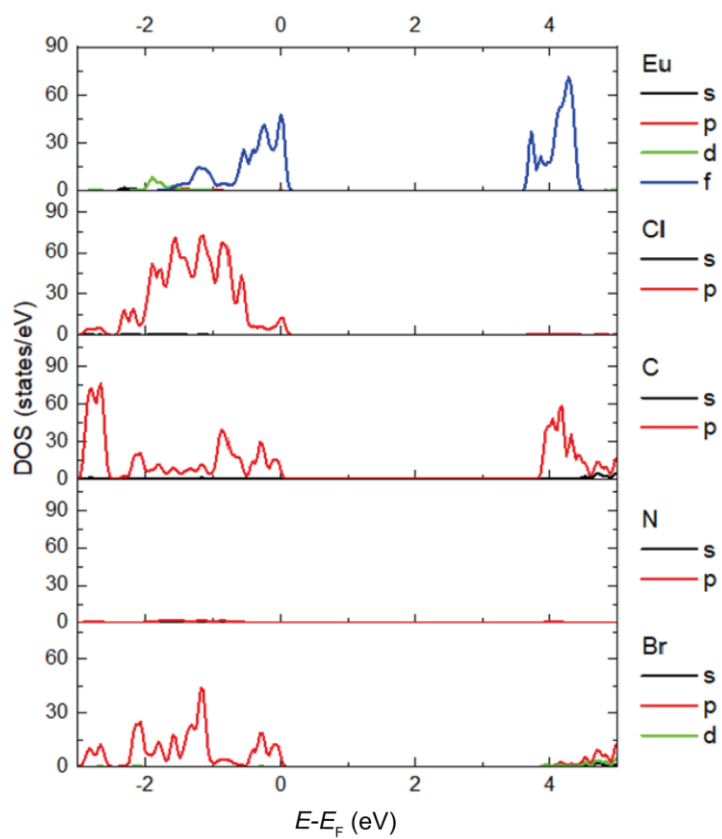
Supplementary Fig. 6 | Thermal stability for chiral europium halides. The thermogravimetric analysis (TGA) curve of $(S-3BrMBA)_3EuCl_6$.



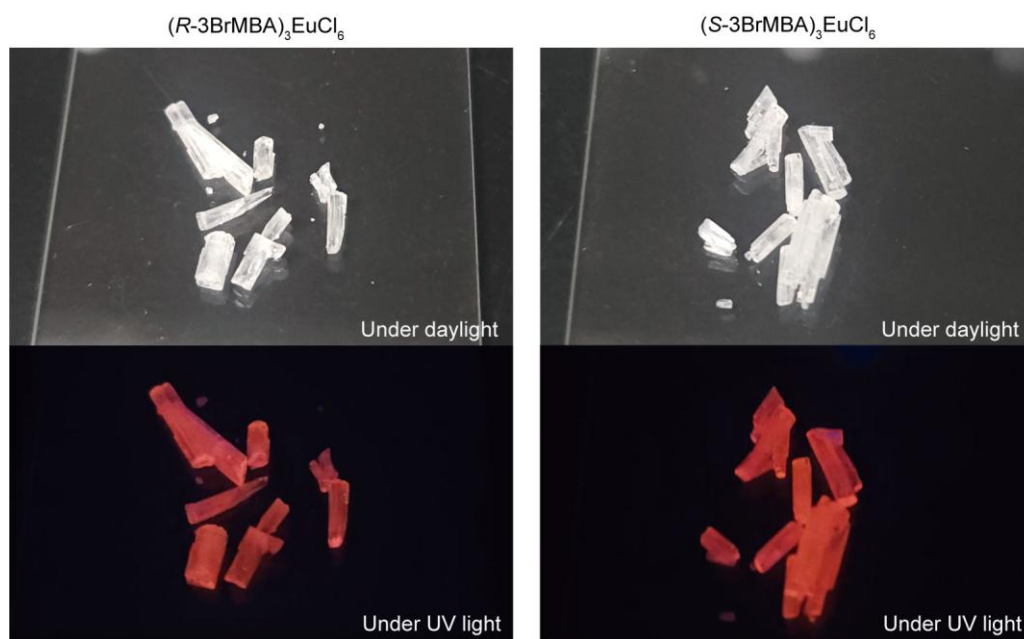
Supplementary Fig. 7 | UV-Vis absorption spectra for chiral europium halides. The normalized UV-Vis absorption spectra of $(R\text{-}3\text{BrMBA})_3\text{EuCl}_6$ (black) and $(S\text{-}3\text{BrMBA})_3\text{EuCl}_6$ (red).



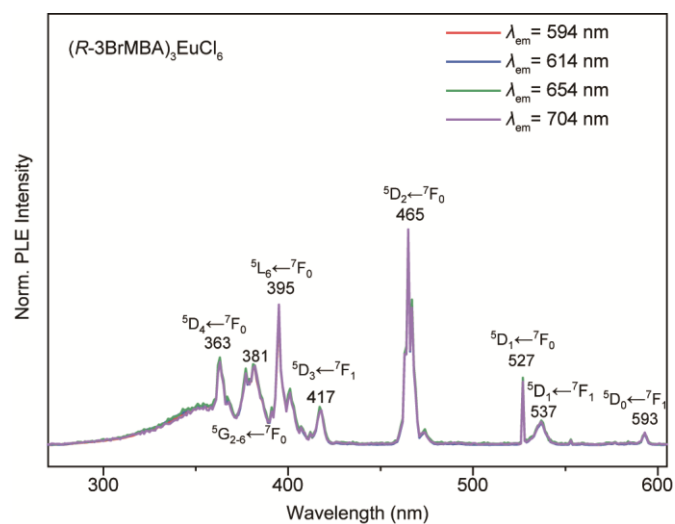
Supplementary Fig. 8 | Absorption spectra and the corresponding circular dichroism (CD) spectra of *R/S*-3BrMBA. (a) The absorption spectra and (b) CD spectra of *R*-3BrMBA (black) and *S*-3BrMBA (red).



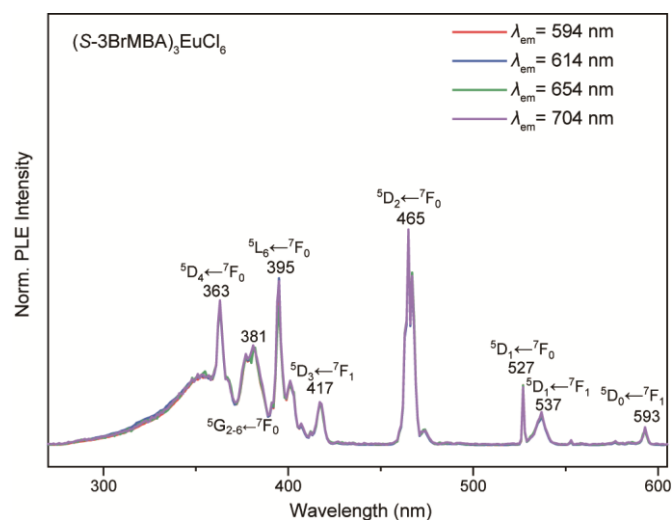
Supplementary Fig. 9 | The projected density of states (PDOS) diagrams of all elements in $(R-3BrMBA)_3EuCl_6$.



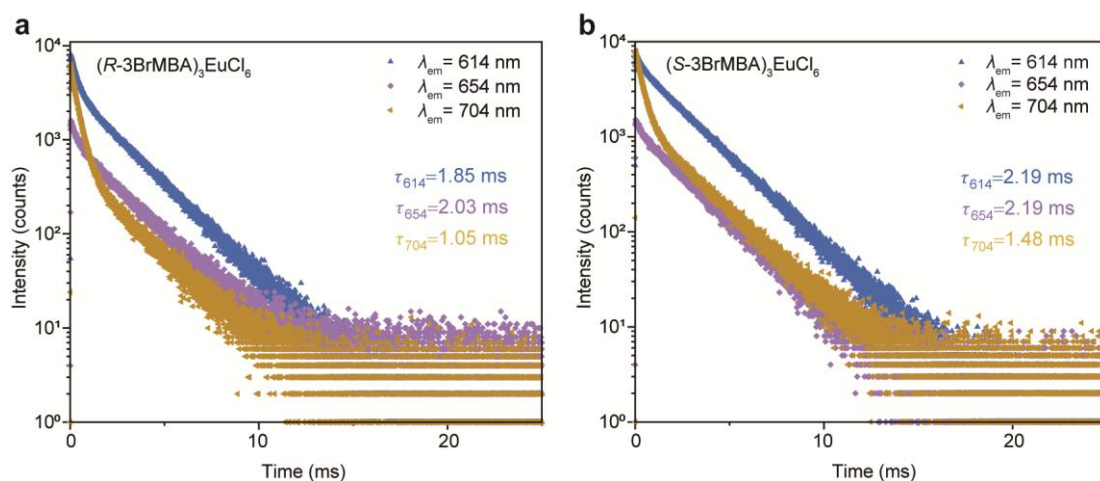
Supplementary Fig. 10 | The (R/S-3BrMBA)₃EuCl₆ crystals under daylight (top) and UV light (bottom).



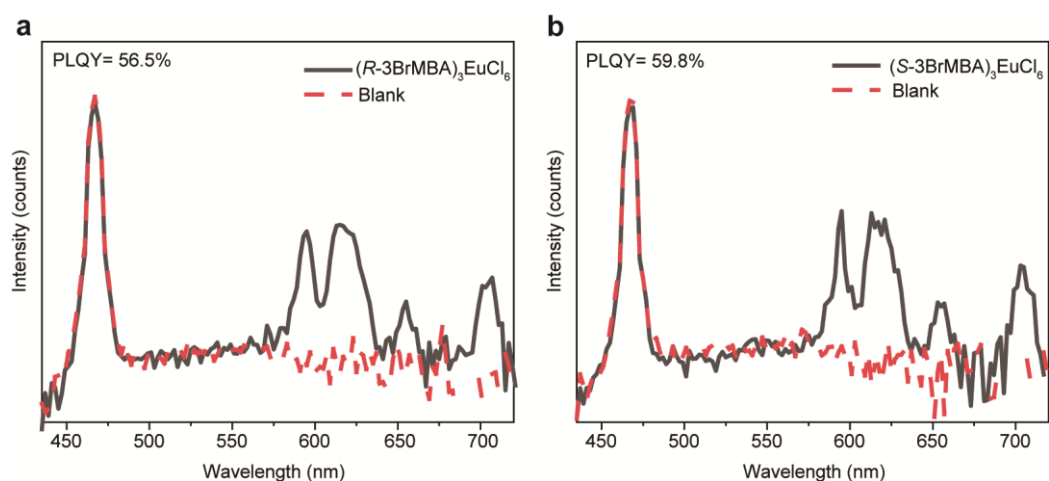
Supplementary Fig. 11 | Photoluminescence excitation (PLE) spectra of $(R-3BrMBA)_3EuCl_6$. The normalized PLE spectra of $(R-3BrMBA)_3EuCl_6$ were recorded under different emission peaks. The red, blue, green, and purple lines represent PLE spectra under different emission peaks at 594, 614, 654 and 704 nm, respectively.



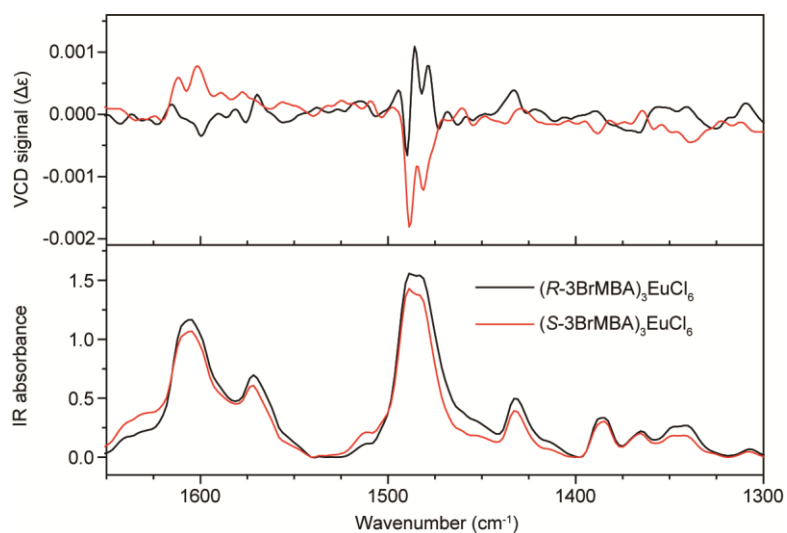
Supplementary Fig. 12 | PLE spectra of (S-3BrMBA)₃EuCl₆. The normalized PLE spectra of (S-3BrMBA)₃EuCl₆ were recorded under different emission peaks. The red, blue, green, and purple lines represent PLE spectra under different emission peaks at 594, 614, 654 and 704 nm, respectively.



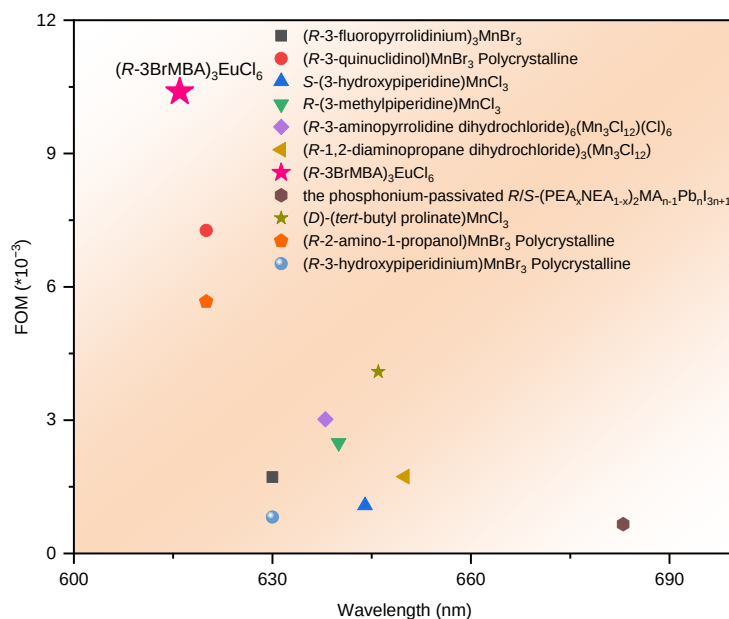
Supplementary Fig. 13 | Time-resolved photoluminescence (TRPL) spectra at different emission peaks for chiral europium halides under 465 nm excitation. The PL lifetime (τ) of $(R-3BrMBA)_3EuCl_6$ (a) and $(S-3BrMBA)_3EuCl_6$ (b) at different emission peaks. The blue, purple and yellow dots represent TRPL spectra under different emission peaks at 614, 654 and 704 nm, respectively.



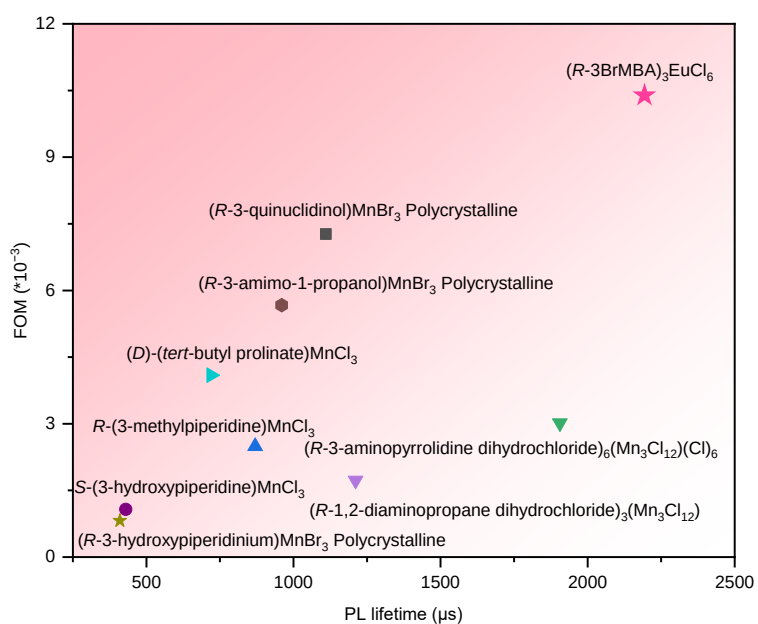
Supplementary Fig. 14 | PLQY measurements for chiral europium halides. The PLQYs of (*R*-3BrMBA)₃EuCl₆ (a) and (*S*-3BrMBA)₃EuCl₆ (b) under 465 nm excitation.



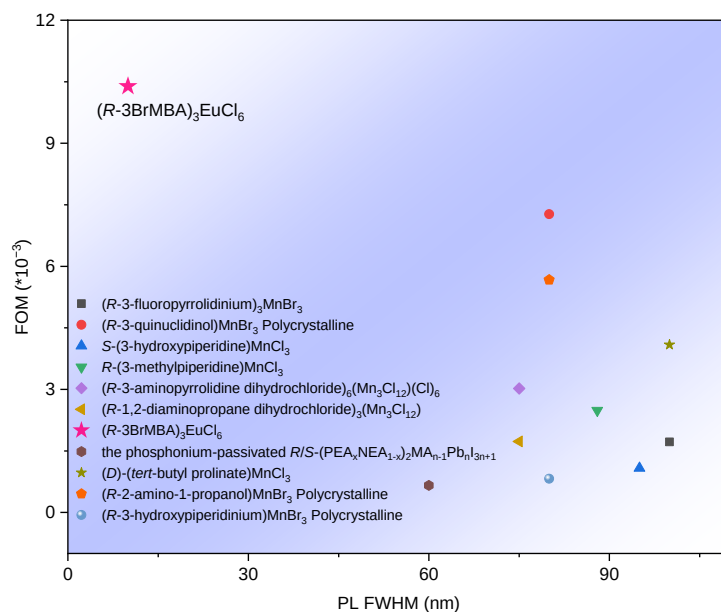
Supplementary Fig. 15 | Vibrational circular dichroism (VCD) spectra and the corresponding infrared (IR) spectra for chiral europium halides. Experimentally measured VCD and IR spectra for (R-3BrMBA)₃EuCl₆ (black) and (S-3BrMBA)₃EuCl₆ (red).



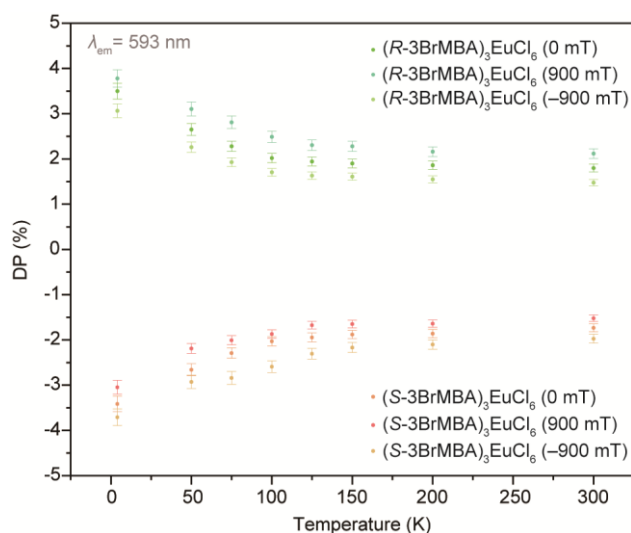
Supplementary Fig. 16 | Efficient red emission of chiral europium halides. Comparison between the emission wavelength and the figure of merit (FOM) of chiral metal halides with red emission reported in literatures and this work.



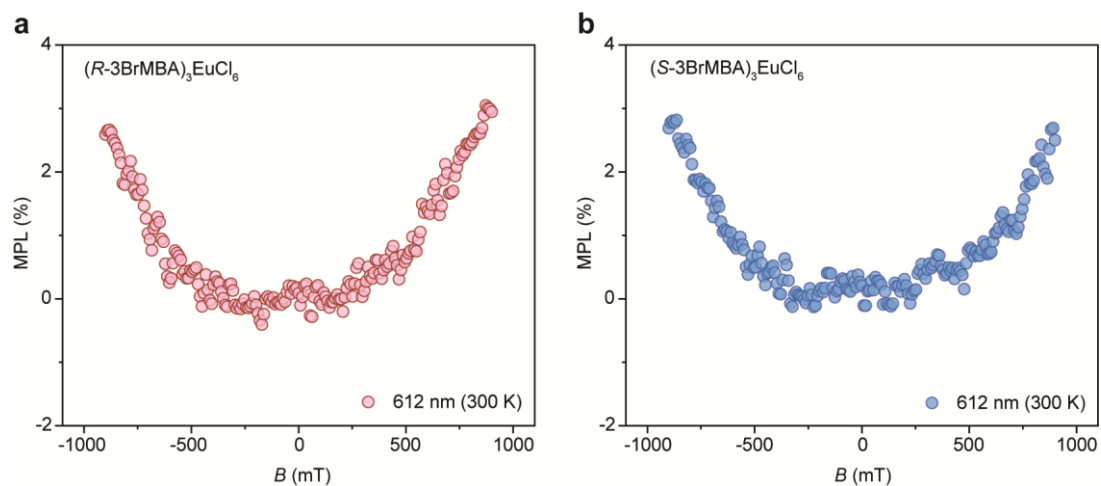
Supplementary Fig. 17 | The long lifetime of chiral europium halides. Comparison between the PL lifetime and the figure of merit (FOM) of chiral metal halides with red emission reported in literatures and this work.



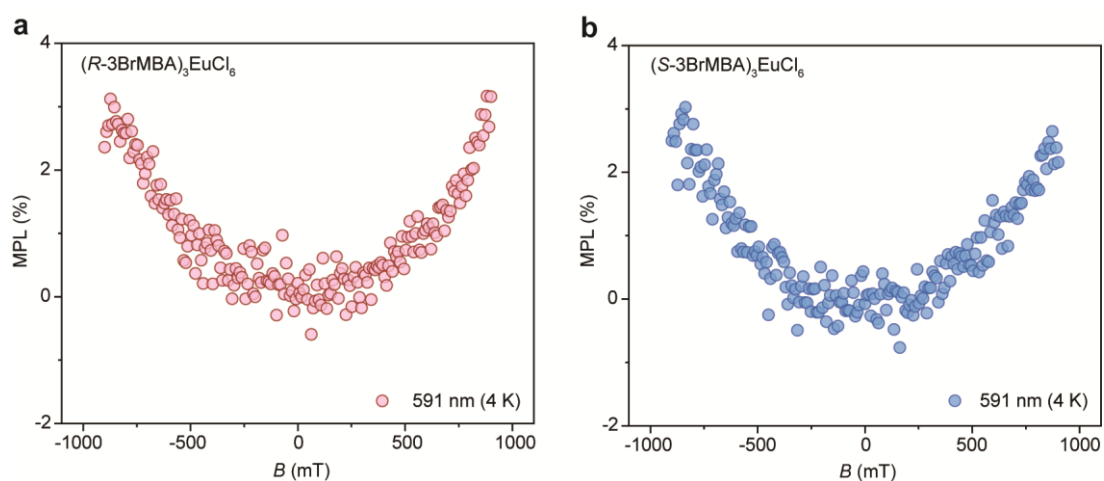
Supplementary Fig. 18 | The narrow bandwidth of chiral europium halides. Comparison between the PL full width at half maximum (FWHM) and the figure of merit (FOM) of chiral metal halides with red emission reported in literatures and this work.



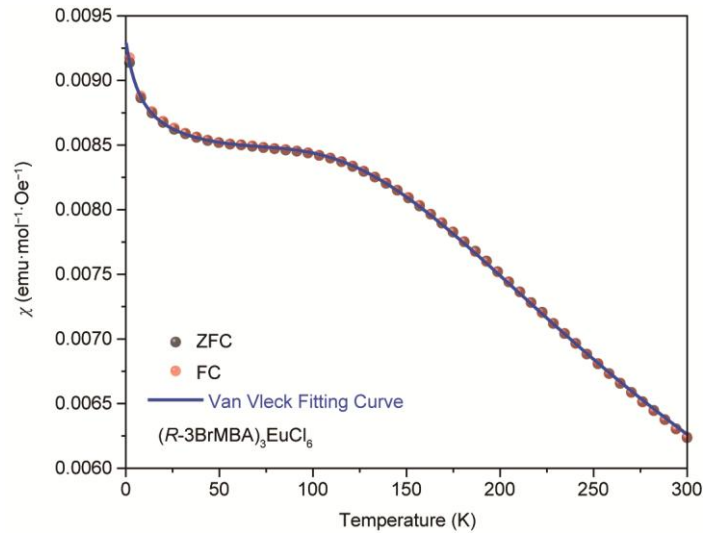
Supplementary Fig. 19 | Experimental results of the temperature-dependent degree of photoluminescence polarization (DP) without and with a constant in-plane magnetic field (± 900 mT). The samples are microcrystals, either $(R-3BrMBA)_3EuCl_6$ or $(S-3BrMBA)_3EuCl_6$. By averaging the results of the three sets of tests, the error bars were processed to include the range of the maximum deviation. The green, blue-green and light green dots represent the temperature-dependent DP under 0 mT, 900 mT and -900 mT for $(R-3BrMBA)_3EuCl_6$, respectively. The orange, pink and yellow dots represent the temperature-dependent DP under 0 mT, 900 mT and -900 mT for $(S-3BrMBA)_3EuCl_6$, respectively.



Supplementary Fig. 20 | Magneto-photoluminescence (MPL) measurements at 612 nm and 300 K. The MPL of $(R-3BrMBA)_3EuCl_6$ (pink) (a) and $(S-3BrMBA)_3EuCl_6$ (blue) (b) at 612 nm and 300 K.



Supplementary Fig. 21 | MPL measurements at 591 nm and 4 K. The MPL of $(R-3BrMBA)_3EuCl_6$ (pink) (a) and $(S-3BrMBA)_3EuCl_6$ (blue) (b) at 591 nm and 4 K.



Supplementary Fig. 22 | Magnetic properties. The temperature dependence of the magnetic susceptibility (χ) curves of $(R-3BrMBA)_3EuCl_6$ in an applied magnetic field of 1000 Oe under the zero-field-cooling (ZFC, black) and field-cooling (FC, red) process, respectively. The blue line represents the fitting curve. The Van Vleck fitted equation is given below:¹

$$\chi = \left(\frac{N\beta}{3kT\chi} \right) \times \frac{24 + \left(\frac{27x}{2} - \frac{2}{3} \right) e^{-x} + \left(\frac{135x}{2} - \frac{5}{2} \right) e^{-3x} + \left(\frac{189x}{2} - \frac{7}{2} \right) e^{-6x} + \left(\frac{405x}{2} - \frac{9}{2} \right) e^{-10x} + \left(\frac{1485x}{2} - \frac{11}{2} \right) e^{-15x}}{1 + 3e^{-x} + 5e^{-3x} + 7e^{-6x} + 9e^{-10x} + 11e^{-15x} + 13e^{-21x}} \quad (1)$$

$$\chi = \frac{\zeta}{kT} \quad (2)$$

The low-temperature region between 2 K to 100 K is affected by the contribution of classical Curie paramagnetic behavior, due to the high magnetic susceptibility caused by low temperature. The gradual decrement in the magnetic susceptibility above 100 K is dominated by Van Vleck paramagnetic behavior, with a coupling constant of about 356.76 cm^{-1} , which is within the reported coupling range of some Van Vleck paramagnetism about Eu compounds ($333 \sim 370 \text{ cm}^{-1}$). At room temperature, the total magnetic behavior (the sum of Curie paramagnetism and Van Vleck paramagnetism) exhibits a magnetic moment of $3.7 \mu_B$.

Supplementary Tables

Supplementary Table 1. The crystal data of (R-3BrMBA)₃EuCl₆ and (S-3BrMBA)₃EuCl₆.

Name	(R-3BrMBA) ₃ EuCl ₆	(S-3BrMBA) ₃ EuCl ₆
Empirical formula	C ₂₄ H ₃₃ Br ₃ N ₃ EuCl ₆	C ₂₄ H ₃₃ Br ₃ N ₃ EuCl ₆
Formula weight (g mol ⁻¹)	967.92	967.92
Temperature (K)	100.00(10)	100.00(10)
Crystal system	Orthorhombic	Orthorhombic
Space group	<i>P</i> 2 ₁ 2 ₁ 2 ₁	<i>P</i> 2 ₁ 2 ₁ 2 ₁
<i>a</i> (Å)	7.6546(2)	7.6510(2)
<i>b</i> (Å)	14.6206(5)	14.6533(4)
<i>c</i> (Å)	31.5842(9)	31.5797(8)
α (°)	90	90
β (°)	90	90
γ (°)	90	90
<i>Z</i>	4	4
Volume (Å ³)	3534.74(18)	3540.48(16)
Flack parameter	-0.010(5)	-0.001(5)

Supplementary Table 2. The parameters of some representative CPL materials with red emission in chiral hybrid metal halides.

	Red-CPL Materials	$ g_{lum} $	PLQY	FOM ($\times 10^{-3}$)	PL FWHM (nm)	PL Lifetime (ms)	Ref.
1	(<i>R</i> -3-fluoropyrrolidinium) ₃ MnBr ₃	0.61%	28.13%	1.72	100	/	2
2	(<i>R</i> -3-quinuclidinol)MnBr ₃ Polycrystalline	2.30%	31.6%	7.27	80	1.11	3
3	(<i>R</i> -2-amino-1-propanol)MnBr ₃ Polycrystalline	1.91%	29.7%	5.67	80	0.96	3
4	(<i>R</i> -3-hydroxypiperidinium)MnBr ₃ Polycrystalline	0.85%	9.7%	0.82	80	0.41	3
5	(<i>D</i>)-(tert-butyl proline)MnCl ₃	0.61%	67.1%	4.09	100	0.72	4
6	<i>S</i> -(3-hydroxypiperidine)MnCl ₃	0.17%	61.13%	1.08	95	0.43	5
7	<i>R</i> -(3-methylpiperidine)MnCl ₃	0.25%	99.70%	2.49	88	0.87	5
8	(<i>R</i> -3-aminopyrrolidine dihydrochloride) ₆ (Mn ₃ Cl ₁₂)(Cl) ₆	0.71%	42.5%	3.02	75	1.906	6
9	(<i>R</i> -1,2-diaminopropane dihydro chloride) ₃ (Mn ₃ Cl ₁₂)	0.2%	86.3%	1.73	75	1.211	6
10	the phosphonium-passivated <i>R/S</i> - (PEA _x NEA _{1-x}) ₂ MA _{<i>n</i>-1} Pb _{<i>n</i>} I _{3<i>n</i>+1}	0.21%	31.5%	0.66	60	/	7
	(<i>R</i> -3BrMBA) ₃ EuCl ₆	1.84%	56.5%	10.40	<10	≈2	This work

Supplementary references

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